

The Impact of Multisession Sleep-Hygiene Strategies on Sleep Parameters in Elite Swimmers

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Purpose: Short sleep duration and poor sleep quality are common in swimmers. Sleep-hygiene strategies demonstrated beneficial effects on several sleep parameters. The present study assessed the impact of a multisession sleep-hygiene training course on sleep in elite swimmers. **Methods:** Twenty-eight elite swimmers (17 [2] y) participated. The sleep-hygiene strategy consisted of 3 interventions. Sleep was measured by actigraphy for 7 days before the beginning of the intervention (baseline), after the first collective intervention (postintervention), after the second collective intervention (postintervention 2), and, finally, after the individual intervention (postintervention 3). The Epworth Sleepiness Scale (ESS) was completed concurrently. Swimmers were classified into 2 groups: nonsomnolent (baseline ESS score ≤ 10 , $n = 13$) and somnolent (baseline ESS score ≥ 11 , $n = 15$). **Results:** All swimmers had a total sleep time of < 8 hours per night. Sixty percent of swimmers were moderately morning type. Later bedtime, less time in bed, and total sleep time were observed in the somnolent group compared with the nonsomnolent group at baseline. An interaction between training course and group factors was observed for bedtime, with a significant advance in bedtime between baseline, postintervention 2, and postintervention 3 for the somnolent group. **Conclusions:** The present study confirms the importance of implementing sleep-hygiene strategies, particularly in athletes with an ESS score ≥ 11 . A conjunction of individual and collective measures (eg, earlier bedtime, napping, and delaying morning training session) could favor the total sleep time achieved.

Keywords: sleepiness, athletes, sleep-hygiene training course

Sleep is considered an essential component of recovery in athletes.^{1,2} The National Sleep Foundation recommends a sleep duration of 8 to 10 hours per day for teenagers (14–17 y) and 7 to 9 hours per day for young adults (18–25 y).³ Studies on sleep restriction/extension have revealed the pivotal role of sleep in cognition, physical performance, mood state, illness, and injury risk.^{4–7} However, insufficient sleep duration and poor sleep quality are common in athletes.^{8,9} Studies involving athletes from different sports showed that total sleep time may be insufficient⁹ and that $> 40\%$ of athletes were identified as poor sleepers.¹⁰

Nonsport factors and sport-specific factors contribute to sleep disturbances in athletes.¹¹ High training load, unfamiliar sleeping environment, early morning training sessions, precompetition anxiety, and long-haul travel are sport-specific constraints that negatively impact sleep quantity and quality.¹¹

Swimmers are particularly exposed to sleep restriction and sleep disturbances due to early morning training sessions (before 8:00 AM) and high training demands.^{12,13} Sleep restriction occurs when individuals are exposed to conditions that reduce their total sleep time below their daily needs.¹⁴ In studies that compared the quality and quantity of sleep in athletes from various sports, the shortest sleep duration⁸ and the highest sleep fragmentation index¹³ were observed in swimmers. Regular short night sleep periods and

impaired sleep quality among athletes, especially swimmers, negatively affect daytime cognitive functioning and increase the level of sleepiness during the day.^{8,15–17} Mild excessive to severe excessive daytime sleepiness (ie, Epworth Sleepiness Scale [ESS] score ≥ 11) was reported in athletes from individual and team sports, with the highest score observed among swimmers (ie, 12).¹³

Sleep training courses are proposed to optimize athletes' sleep quality and quantity¹⁸ and deliver techniques for improved sleep hygiene, assisted sleep, and sleep extension.¹⁸ Sleep hygiene strategy is defined as a set of recommendations intended to promote healthy sleep by advising on how to deal with lifestyle, behavioral, and environmental factors.¹⁹ Sleep hygiene strategy usually focuses on sleep/wake cycles, prebedtime routines, the bedroom environment, electronic device-induced light exposure, and caffeine consumption.^{20–24}

A systematic review and meta-analysis demonstrated the benefits of single-session and multisession sleep hygiene strategy in athletes¹⁸ with benefits on sleep quality,²⁵ sleep-onset variability,²⁵ sleep efficiency,²⁵ and sleep duration.²² In addition, studies have reported beneficial effects on sport-specific performance^{21,26} and mood.²⁷ However, some limitations of these studies must be highlighted. The number and duration of the sessions as well as the time elapsed between sessions differ across studies. We currently do not know whether the benefits persist over time.^{20,21,25} To the best of our knowledge, no study has assessed the interest of sleep hygiene strategy among swimmers.

The present study aimed to assess the effects of a multisession sleep hygiene training course for sleep improvement in elite swimmers. We hypothesized that a sleep hygiene training course may improve the total sleep time and sleep quality of swimmers, especially those with a high level of sleepiness.

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Methods

Subjects

A total of 28 swimmers (12 women, 16 men, age = 17 [2] y), recruited from 3 training centers, participated in this study.

All included swimmers were injury free, did not present with any sleep or psychiatric disorders, participated in all sleep hygiene training courses, and wore the actigraphic device during all the training courses. Sleep data were available for a total of 728 nights (Figure 1). Two swimmers were considered as tier 4 (elite/international level), and 26 swimmers were tier 3 (highly trained/national level) according to the classification proposed by McKay et al.²⁸ Data were collected during the regular season apart from competition periods. All swimmers trained twice daily during the week (ie, morning and afternoon) and once during the weekend (ie, Saturday morning) for a weekly training volume of ≈ 20 hours.

The National Ethics Committee for the Research in Sciences and Technics of Physical Activity and Sports approved the protocol (IRB00012476-2020-30-11-71). This study was conducted in accordance with current national and international laws and regulations governing the use of human subjects (Declaration of Helsinki II). All participants over 18 years of age provided written informed consent after receiving complete information about the present study. Agreement from the parents was obtained for minors.

Design

The sleep hygiene training course included 3 sessions provided by 2 researchers (F.P.—PhD and F.S.—MD, PhD). Each session was separated by 1 month. The protocol was carried out from November 15, 2020, to March 15, 2021. All participants received the same theoretical content during interventions. The first session of the sleep hygiene training course was collective and focused on the role of sleep in recovery,² sleep quantity recommendations,³ negative impacts of sleep deprivation on cognitive and physical performance, sleep measurement, and the role of the nap to extending sleep quantity.^{29,30} The second collective training course started with a presentation (mean [SD]) of the sleep data of the group studied (ie, bedtime, time in bed, total sleep time, and the number of naps). Then, the second session continued with a presentation of possible countermeasures to improve sleep quantity and quality: nap, sleep extension,⁶ banking sleep strategies,³¹ and finally advice on food and caffeine consumption. Advice on the importance of maintaining a regular wake/sleep cycle during workdays and free days (ie, avoiding social jet lag³²) was also addressed during the second sleep hygiene training course. Concepts and models of sleep regulation proposed by Borbély et al³³ were discussed during this session. The first and second sessions lasted 1:30, and a short nap (00:30) was taken on a mat as a practical training³⁰ after the presentation. During the last training course, each participant had a face-to-face presentation of their own sleep data. Advice was proposed according to bedtime and wake-up time variability, total sleep time, sleep efficiency, number and duration of naps per week, and daytime sleepiness. If an irregular wake/sleep cycle during workdays and free days was observed, the detrimental impacts of social jet lag were presented, and the participants were reminded of the importance of maintaining a regular wake/sleep cycle.

Sleep measures were carried out for 7 days before the beginning of the training course program (baseline), after the first collective training course (postintervention), after the second

collective training course (postintervention 2), and after the individual training course (postintervention 3). The study design is presented in Figure 1.

Measures

Participants' chronotype was assessed using the Morningness–Eveningness Questionnaire.³⁴ According to the score obtained, individuals were classified as definitely morning type (score > 70), moderately morning type (59 < score < 69), neither type (42 < score < 58), moderately evening type (31 < score < 41), or definitely evening type (30 < score).

Participants were required to continuously wear an actigraphic device (Motion Watch 8, Neurotechnology) on the nondominant wrist during the measurement period, except when swimming or showering. Acceleration above 0.01 G threshold was detected by a piezoelectric accelerometer. Activity data were stored and summed over a 1-minute epoch. Self-reported sleep diaries were completed in conjunction with wrist activity monitors. Bedtime, wake-up time, night awakenings, and nap duration (in minutes) were manually reported by participants every day. Motionwatch MotionWare software (Cambridge Neurotechnology Ltd) was used to analyze sleep data. The following variables were derived from the sleep analysis function:

- Bedtime (h:min): the time at which a subject goes to bed for sleep
- Wake-up time (h:min): the time at which a subject gets out of bed
- Time in bed (h:min): total time spent in bed
- Sleep latency (h:min): time taken from bedtime to sleep onset
- Wake after sleep onset (WASO; in minutes): total time spent awake during the sleep period
- Total sleep time (h:min): total time spent asleep
- Sleep efficiency (in percentage): (total sleep time divided by time in bed) $\times 100$
- Number of naps (arbitrary unit: au): sum of naps taken during the week
- Nap duration: mean nap duration for napping participants

Daytime sleepiness was measured using the ESS,³⁵ and the sleep quality using the Pittsburgh Sleep Quality Index (PSQI).³⁶ The ESS is composed of 8 items, which refer to everyday situations. For each item, participants had to rate their level of sleep ability. The global score varied from 0 to 24 au. ESS is commonly used to identify individuals with excessive daytime sleepiness (ESS score ≥ 11 au). In the present study, participants with a global ESS score ≥ 11 au were assigned to a somnolent group, and participants with a score ≤ 10 au to a nonsomnolent group.

The PSQI consists of 19 items estimating 7 components of sleep quality: subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disruptions, the use of sleep medications, and daytime dysfunction. After completion, participants obtained a global score between 0 and 21 au. A global score > 5 au indicated poor sleep quality.³⁶ The ESS and the PSQI were completed at 18:00 at the end of each measurement period.

Statistical Analysis

Statistical analysis was conducted using R software (version 4.1.2). To compare sleep data at baseline, a Student *t* test comparison

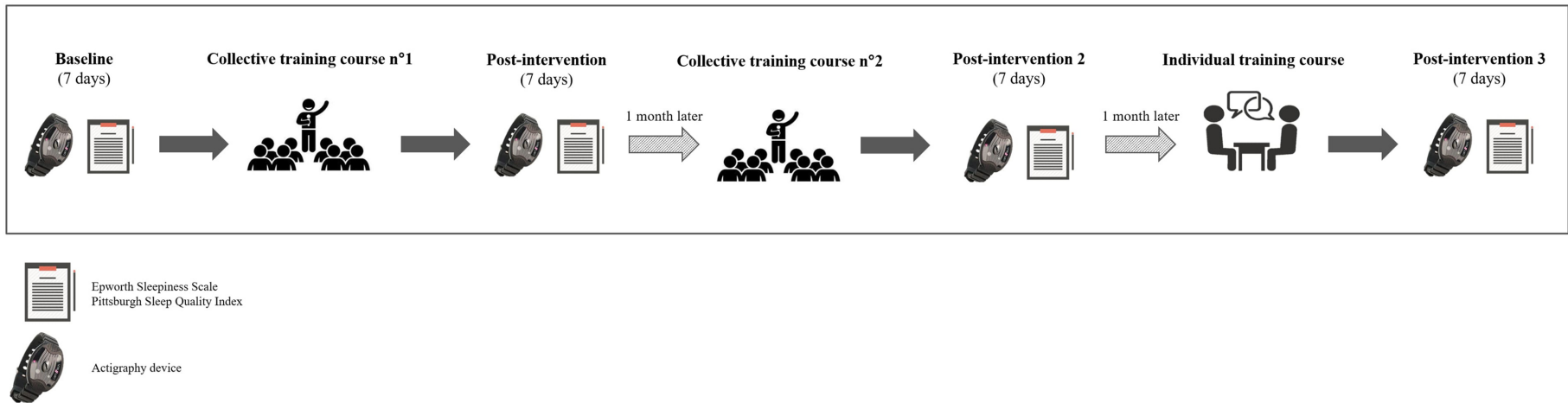


Figure 1 — Study design. *Note.* Sleep data were available for a total of 728 nights. Number of nights = (28 swimmers × 7 nights × 4 measurement periods) – missing nights.

between groups (somnolent and nonsomnolent) was applied. The normality of the distribution was assessed with the Shapiro–Wilk test and the equality of variances with the Fisher test. The standardized Cohen (*d*) effect size (ES) analysis was used to interpret the magnitude of the mean differences between groups. An ES was considered to be trivial (<0.20), small ($0.20–0.49$), medium ($0.50–0.79$), or large (>0.80).³⁷

Sleep variables' intraindividual means were calculated for each sleep parameter by averaging the individual's values measured during the 7 nights after each training course. The intraindividual standard deviations for sleep variables were computed to indicate the variation from the participant's mean during the 7 days of measurement.³⁸ The normality of the distribution was assessed with the Shapiro–Wilk test, and the sphericity (ie, homogeneity of variance and covariance) was assessed with the Mauchly test. Degrees of freedom were adjusted according to the procedure of Greenhouse–Geisser when the assumption of sphericity was not met. Mixed analysis of variance with 1 repeated factor (training course) and 1 independent factor (group) was applied on sleep parameters. If a significant interaction between factors was observed, a paired comparison with Bonferroni correction was performed. Results are presented as mean (SD). $P < .05$ was considered significant for all statistical analyses.

Results

Sleep Characteristics at Baseline

The Morningness–Eveningness Questionnaire revealed that swimmers were either moderately morning type ($n = 17$) or neither type ($n = 11$). Analysis of ESS score distribution demonstrated that 13 swimmers had a score ≤ 10 au (ie, lower to higher normal daytime sleepiness), and 15 swimmers obtained a score ≥ 11 au (ie, mild to severe daytime sleepiness). All sleep data are presented in Table 1. Significant differences between groups were observed for bedtime ($P = .04$, $ES = 0.84$), time in bed ($P = .04$, $ES = 0.79$), total sleep time ($P = .03$, $ES = 0.85$), number of naps ($P < .01$, $ES = 1.07$), and PSQI global score ($P < .001$, $ES = 1.09$). No significant difference between groups was reported for other sleep parameters.

Mean Sleep Data Analysis

Statistical analysis demonstrated a significant group effect on number of naps per week ($P = .02$) and on ESS score ($P < .001$). A significant training course effect was observed on bedtime ($P < .001$), wake-up time ($P < .001$), time in bed ($P < .01$), sleep efficiency ($P = .03$), and WASO ($P < .01$).

The analysis of variance revealed a significant interaction between training course and group factors for bedtime ($P = .02$). Post hoc analysis revealed a significantly advanced bedtime between baseline (22:54 [00:46]) and postintervention 2 (22:24 [00:39]; $P < .001$, $ES = 0.73$) and postintervention 3 measures (22:30 [00:38]; $P = .04$, $ES = 0.59$) for the somnolent group (Table 2). No significant interaction between factors was observed for the following variables: wake-up time ($P = .41$), time in bed ($P = .13$), total sleep time ($P = .23$), sleep latency ($P = .87$), sleep efficiency ($P = .91$), WASO ($P = .67$), number of naps ($P = .21$), nap duration ($P = .15$), ESS score ($P = .50$), and PSQI global score ($P = .16$).

Sleep Variability Analysis

The analysis of variance demonstrated a significant training course effect on wake-up time variability ($P = .03$) and a significant group effect on total sleep time variability ($P = .02$; Table 3). The statistical analysis did not show any significant interaction between group and training course factors for the following sleep parameters: bedtime ($P = .24$), wake-up time ($P = .89$), time in bed ($P = .54$), total sleep time ($P = .58$), sleep latency ($P = .24$), sleep efficiency ($P = .35$), and WASO ($P = .34$; Table 3).

Discussion

The present study assessed the interest of a multisession sleep hygiene strategy on the quantity and quality of sleep in elite swimmers. Results showed a positive effect of the sleep hygiene training course with earlier bedtimes, especially among the somnolent group.

Sleep data analysis revealed that the average swimmer goes to bed at 22:42 (00:43), falls asleep after 20 minutes, wakes up at

Table 1 Sleep Parameters at Baseline

	Total, N = 28	Nonsomnolent group, n = 13	Somnolent group, n = 15	P	Effect size
Bedtime, h:min	22:42 (00:43)	22:24 (00:30)	22:54 (00:46)	.04	0.84
Wake-up time, h:min	06:51 (00:45)	06:48 (00:42)	06:53 (00:49)	.77	0.11
Time in bed, h:min	08:09 (00:38)	08:25 (00:33)	07:56 (00:38)	.04	0.79
Total sleep time, h:min	06:41 (00:46)	07:00 (00:51)	06:24 (00:33)	.03	0.85
Sleep latency, h:min	00:20 (00:12)	00:23 (00:16)	00:18 (00:08)	.30	0.40
Sleep efficiency, %	81.80 (5.73)	83.00 (5.99)	80.80 (5.47)	.31	0.39
WASO, h:min	01:04 (00:23)	00:57 (00:17)	01:10 (00:26)	.14	0.58
Number of naps per week, au	1.93 (2.12)	0.85 (1.77)	2.87 (2.0)	<.01	1.07
Nap duration per week, h:min	01:19 (00:29)	01:27 (00:28)	01:16 (00:31)	.56	0.34
ESS score, au	10.6 (4.60)	6.54 (2.70)	14.10 (2.52)	<.001	2.36
PSQI score, au	4.11 (2.39)	3.00 (1.13)	5.00 (2.78)	<.001	1.09

Abbreviations: ESS, Epworth Sleepiness Scale; PSQI, Pittsburgh Sleep Quality Index; WASO, wake after sleep onset. Note: Effect size was considered to be trivial (<0.20), small ($0.20–0.49$), medium ($0.50–0.79$), or large (>0.80).³⁷

Note: Bold indicates significant values ($P < .05$).

Table 2 Sleep Characteristics at Baseline and After the First Collective Training Course (PI), Second Collective Training Course (PI 2), and Individual Training Course (PI 3) for Nonsomnolent and Somnolent Groups

	Nonsomnolent group, n = 13				Somnolent group, n = 15				Group effect	Training course effect	Group × training course effect
	Baseline	PI	PI 2	PI 3	Baseline	PI	PI 2	PI 3			
Bedtime, hh:min	22:24 (00:30)	22:30 (00:36)	22:18 (00:30)	22:18 (00:31)	22:54 (00:46)	22:54 (00:50)	22:24 (00:39) ^{a,***}	22:30 (00:38) ^{a,*}	.19	<.001	.02
Wake-up time, hh:min	06:48 (00:42)	06:48 (00:27)	06:52 (00:27)	06:25 (00:32)	06:53 (00:49)	07:07 (00:52)	06:51 (00:40)	06:41 (00:34)	.39	<.001	.41
Time in bed, hh:min	08:25 (00:33)	08:18 (00:28)	08:32 (00:37)	08:05 (00:29)	07:56 (00:38)	08:12 (00:34)	08:26 (00:23)	08:10 (00:36)	.48	<.01	.13
Total sleep time, hh:min	07:00 (00:51)	06:57 (00:41)	06:58 (00:39)	06:42 (00:37)	06:24 (00:33)	06:35 (00:25)	06:36 (00:25)	06:28 (00:44)	.09	.15	.23
Sleep latency, hh:min	00:23 (00:16)	00:22 (00:14)	00:22 (00:15)	00:20 (00:15)	00:18 (00:08)	00:16 (00:12)	00:19 (00:10)	00:18 (00:11)	.39	.92	.87
Sleep efficiency, %	83.00 (5.99)	83.90 (6.30)	81.80 (5.87)	82.90 (5.04)	80.80 (5.47)	80.60 (6.06)	78.40 (5.70)	79.10 (6.28)	.14	.03	.91
WASO, hh:min	00:57 (00:17)	00:54 (00:21)	01:08 (00:22)	00:58 (00:14)	01:10 (00:26)	01:16 (00:36)	01:26 (00:30)	01:19 (00:26)	.05	<.01	.67
Number of naps per week, au	0.85 (1.77)	0.69 (1.70)	0.62 (1.26)	0.77 (1.30)	2.87 (2.00)	1.67 (1.95)	2.60 (2.56)	1.73 (1.83)	.02	.17	.21
Nap duration per week, hh:min	01:27 (00:28)	01:09 (00:15)	01:25 (00:43)	01:43 (00:38)	01:16 (00:31)	01:17 (00:15)	01:02 (00:24)	01:15 (00:32)	.35	.29	.15
ESS score, au	6.54 (2.70)	6.00 (3.79)	4.62 (3.33)	5.20 (3.33)	14.10 (2.52)	13.10 (3.42)	11.70 (4.43)	11.40 (5.14)	<.001	.05	.50
PSQI score, au	3.00 (1.13)	3.08 (1.85)	3.46 (1.85)	3.50 (2.59)	5.00 (2.78)	5.07 (2.06)	4.21 (2.58)	4.36 (2.31)	.08	.98	.16

Abbreviations: PI, postintervention; ESS, Epworth Sleepiness Scale; PSQI, Pittsburgh Sleep Quality Index; WASO, wake after sleep onset.

^aSignificant difference compared with baseline (group × training course interaction).* $P < .05$. *** $P < .001$.Note: Bold indicates significant values ($P < .05$).

Table 3 Variability of Sleep Parameters at Baseline, After the First Collective Training Course (PI), Second Collective Training Course (PI 2), and Individual Training Course (PI 3) for the Nonsomnolent and Somnolent Groups

	Nonsomnolent group (n = 13)				Somnolent group (n = 15)				Group effect	Training course effect	Group × training course effect
	Baseline	PI	PI 2	PI 3	Baseline	PI	PI 2	PI 3			
Bedtime, hh:min	00:39 (00:22)	00:30 (00:12)	00:29 (00:16)	00:32 (00:13)	00:45 (00:24)	00:45 (00:16)	00:37 (00:27)	00:29 (00:18)	.31	.08	.24
Wake-up time, hh:min	01:24 (00:28)	01:22 (00:34)	01:13 (00:18)	01:03 (00:31)	01:15 (00:23)	01:14 (00:34)	00:58 (00:17)	01:03 (00:27)	.22	.03	.89
Time in bed, hh:min	01:01 (00:39)	01:08 (00:31)	01:07 (00:16)	00:49 (00:22)	00:51 (00:22)	00:58 (00:29)	00:46 (00:15)	00:48 (00:22)	.09	.11	.54
Total sleep time, hh:min	01:00 (00:39)	01:00 (00:27)	00:58 (00:15)	00:49 (00:18)	00:43 (00:18)	00:47 (00:15)	00:40 (00:14)	00:46 (00:28)	.02	.42	.58
Sleep latency, hh:min	00:17 (00:14)	00:20 (00:11)	00:15 (00:10)	00:17 (00:10)	00:16 (00:05)	00:15 (00:09)	00:18 (00:08)	00:14 (00:07)	.63	.85	.24
Sleep efficiency, %	4.55 (2.85)	4.69 (2.32)	3.86 (1.63)	4.54 (2.21)	3.82 (1.73)	3.86 (1.20)	4.62 (1.58)	5.18 (3.34)	.68	.95	.35
WASO, hh:min	00:14 (00:09)	00:15 (00:08)	00:18 (00:07)	00:14 (00:06)	00:20 (00:09)	00:18 (00:08)	00:19 (00:07)	00:22 (00:15)	.08	.86	.34

Abbreviations: PI, postintervention; WASO, wake after sleep onset.
Note: Bold indicates significant values (*P* < .05).

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06:51 (00:45), and has a total sleep time of 06:41 (00:46) and a sleep efficiency of $\approx 82\%$ at baseline. Total sleep time and efficiency are lower than the National Sleep Foundation recommendations, that is, 7 hours per day and 85% for young adults.^{3,8,9} The Morningness–Eveningness Questionnaire revealed that 60% of the swimmers are moderately morning types. This result is consistent with the study of Lastella³⁹ that showed a significant relationship between sport and chronotype with a higher proportion of morning types involved in disciplines that required early morning training sessions (ie, before 8:00 AM). A previous study demonstrated the detrimental effect of early morning training sessions among adolescent swimmers with a decrease in total sleep time the night before.³⁹ On the contrary, sleep duration is extended on the night preceding a later training session ($\geq 7:00$) or an off day.⁴⁰

In comparison with other disciplines, swimmers demonstrate a high level of diurnal sleepiness, which may be explained by an insufficient amount and/or quality of sleep.¹³ Cognitive (eg, memory, learning capacities, and attention) and physical performance may be impaired when athletes have insufficient sleep time and/or poor sleep quality.^{2,41,42} In the present study, the negative impact of somnolence on academic and sport performance was explained to the swimmers during the training course. The swimmers involved were classified into 2 groups based on their ESS baseline score: the nonsomnolent group (ESS score ≤ 10 au, $n = 13$) and the somnolent group (ESS score ≥ 11 au, $n = 15$). Later bedtime, less time in bed, and less total sleep time were observed in the somnolent group compared with the nonsomnolent group at baseline. Also, a baseline PSQI score >5 indicating low sleep quality was reported by the somnolent group.

A significant interaction between training course and group factors was observed for bedtime. The swimmers in the somnolent group advanced their bedtime by almost 30 minutes from baseline to postintervention 2 and postintervention 3. The present results confirm the importance of implementing sleep hygiene strategies, particularly in athletes with shorter total sleep times, as previously demonstrated in professional rugby league athletes.²⁰ Earlier bedtimes observed in the somnolent group after the second and the third training course confirm the importance of a 2-month multi-session sleep hygiene training course and the impact of explaining and presenting the participants' sleep data. As the total sleep time in the somnolent group did not change significantly over the course of the study, one may speculate that earlier bedtimes led to a progressively more restorative sleep.⁴³ Also, a significant training course effect in wake-up time variability was observed with a decrease for both groups, which may be explained by the fact that the importance of maintaining a regular wake/sleep cycle between work days and free days was presented during the training course.

The ESS score of the somnolent group decreased from 14.1 (2.5) au at baseline to 11.4 (5.1) au at postintervention 3 (group effect, $P < .001$), which is another positive result of the current sleep hygiene strategy.

Napping seems a promising strategy to increase the overall sleep time of young adults.³ In the present study, the swimmers of the somnolent group napped more frequently but for a shorter time than those belonging to the nonsomnolent group. It has been proposed that daytime napping is an effective strategy to supplement nighttime sleep in the context of sleep restriction.⁴⁴ Long-duration naps (ie, 90 min) are particularly effective to improve physical and cognitive performance among athletes experiencing sleep restriction.³⁰ Future studies are required to

find a compromise between napping frequency and duration among athletes.

Practical Applications

At an individual level, earlier bedtimes and naps constitute practical applications for swimmers to decrease the level of diurnal sleepiness. At a collective level, delaying the start of morning training sessions, in conjunction with earlier bedtimes, should increase swimmers' total sleep time.⁴⁵

Limits

Even though the present study demonstrated the beneficial impact of a sleep hygiene strategy among swimmers, some limitations need to be considered. Sleep parameters were measured objectively by an actigraphy device, but there is a possibility that swimmers modified their behaviors under the influence of social desirability.⁴⁶ Actigraphy devices measure sleep parameters based on wrist acceleration but do not allow any insight into sleep architecture.

Sleep hygiene sessions were conducted from November 2020 to March 2021, and it is possible that light exposure variations over this period may have influenced the sleep results (eg, time in bed and total sleep time).

The sleep hygiene training course and measurement periods were implemented during regular training periods according to the coaches. However, the training load was not objectively recorded, which may be considered a limitation of the present results.

Conclusions

The present study highlights the interest of implementing a multi-session sleep-hygiene training course, especially for swimmers with a high level of sleepiness. The sleep-hygiene strategy induced earlier bedtime in the somnolent group, whereas the total sleep time remained constant. Sleep quantity obtained by the swimmers throughout the follow-up period was lower than the National Sleep Foundation recommendations.

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